CS & AI · EXPLAINED

What is an algorithm?

You already use them every day — without even knowing it.

THE HOOK

Made 2-minute Maggi today? You just ran an algorithm.

Boil water → add masala → add Maggi → wait 2 min. **Steps, in order.**

That's all an algorithm is — a recipe. Let's make it click. 🍜

TODAY'S QUESTION

What exactly **is** an algorithm — and why does CS care so **much** about it?

IDEA 1 / 4 — WHAT IT IS

It's just a step-by-step recipe.

```
# algorithm: make 1 cup chai
1. boil 1 cup water
2. add 1 tsp tea + sugar
3. add milk, boil 2 min
4. strain into cup   → done ✓
```

- A clear set of steps...
- ...to solve **one** specific problem.
- That's it. That's an algorithm.

Steps must be clear, ordered & finite.

- **Clear:** no "add thoda sa" — how much, exactly?
- **Ordered:** add Maggi *before* boiling water? Disaster. Order matters.
- **Finite:** it must *end* — no running forever.

A computer follows steps **literally**. Be vague, and it breaks.

Same task → a smarter algorithm is way faster.

Find a word in a **1000-page** dictionary:

checks needed to find it

- **Page by page** = up to 1000 looks.
- **Open the middle**, go left or right, repeat = ~10 looks.
- Both correct — one is 100× faster.

Simple rules, like a queue.

IN → 🥟 1st 🥟 2nd 🥟 3rd → OUT

- Samosa line: whoever comes **first**, gets served **first**.
- That's a **queue** — FIFO (First In, First Out).
- Print jobs, ticket counters, WhatsApp messages — all run on a queue.

WHY IT MATTERS

Algorithms run your whole day.

- **Google Maps** picks the shortest route home.
- **YouTube** decides which video to show you next.
- **UPI** checks and clears a payment in 2 seconds.

All of it = clever step-by-step algorithms, running millions of times a second.

✗ MYTH

"Algorithm = scary maths that only coders and AI use."

✓ REALITY

It's just **clear steps** to do a task — you use them daily. Code is only writing those steps precisely enough for a computer.

RECAP — 2 THINGS TO REMEMBER

1. An algorithm = precise, ordered, finite steps to solve a problem.
2. The same task can have a fast or slow algorithm — that’s the heart of CS.

Algorithms = recipes. Simple. 🍜

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Next question: **How does the internet reach your phone?** 🌐